Practical_No.02

1. **Write a shell script to generate mark- sheet of a student. Take 3 subjects, calculate and display total marks, percentage and Class obtained by the student. Output 1 :**

```
ASUS@LAPTOP-I4P5FEVA MINGW64 ~/OneDrive/Desktop/cd24021--os (main)
$ echo "Enter marks of English"
read m1
echo "Enter marks of Maths"
read m2
echo "Enter marks of Science"
read m3
total=$(( $m1+$m2+$m3 ))
percentage=$(($total/3))
echo "Student: Total Marks = $total"
echo "Percentage = $percentage"
if [ $percentage -ge 75 ]; then
        echo "Class: Distinction"
elif [ $percentage -ge 60 ]; then
        echo "Class: First Class"
elif [ $percentage -ge 40 ]; then
        echo "Class: Second Class"
elif [ $percentage -ge 35 ]; then
        echo "Class: Third Class"
else
        echo "Class: Fail"
fi
Enter marks of English
60
Enter marks of Maths
60
Enter marks of Science
60
Student: Total Marks = 180
Percentage = 60
Class: First Class

ASUS@LAPTOP-I4P5FEVA MINGW64 ~/OneDrive/Desktop/cd24021--os (main)
$
```

2.Write a menu driven shell script which will print the following menu and execute the given task.

- **Display calendar of given task.**
- **Display today's date and time.**
- **Display username those are currently logged in the system.**
- **Display your terminal number Output 2 :**

```
ASUS@LAPTOP-I4P5FEVA MINGW64 ~/OneDrive/Desktop/cd24021--os (main)
$ echo "1. Calendar of current month"
echo "2. Today's date and time"
echo "3. Logged in users"
echo "4. Terminal number"
echo "Enter your choice"
read ch

if [ $ch -eq 1 ]; then
        date +%B %Y"
elif [ $ch -eq 2 ]; then
        date
elif [ $ch -eq 3 ]; then
        who
elif [ $ch -eq 4 ]; then
        tty
else
        echo "Invalid choice"
fi

1. Calendar of current month
2. Today's date and time
3. Logged in users
4. Terminal number
Enter your choice
2
Sat Jan 24 18:58:56 IST 2026
bash: 1.: command not found
1. Calendar of current month
2. Today's date and time
3. Logged in users
4. Terminal number
Enter your choice
2
> Sat jan 24 18:58 IST 2026
> bash:1 commans not found
```

3.Write a shell script which will generate first n Fibonacci numbers like: 1, 1, 2, 3, 5, 13

Output 3 :

```
ASUS@LAPTOP-I4P5FEVA MINGW64 ~/OneDrive/Desktop/cd24021--os (main)
$ #!/bin/bash
echo "Enter how many Fibonacci numbers you want"
read n
a=1
b=1
echo "Fibonacci Series:"
if [ "$n" -ge 1 ]; then
printf "%d " "$a"
fi
if [ "$n" -ge 2 ]; then
printf "%d " "$b"
fi
for (( i=3; i<=n; i++ ))
do
c=$((a + b))
printf "%d " "$c"
a=$b
b=$c
done
Enter how many Fibonacci numbers you want
8
Fibonacci Series:
1 1 2 3 5 8 13 21
ASUS@LAPTOP-I4P5FEVA MINGW64 ~/OneDrive/Desktop/cd24021--os (main)
$
```

4. Write a shell script which will accept a number b and display first n prime numbers as output.

Output 4 :

```
1 1 2 3 5 8 13 21
ASUS@LAPTOP-I4P5FEVA MINGW64 ~/OneDrive/Desktop/cd24021--os (main)
$ #!/bin/bash
echo "Enter the value of n"
read n
count=0
num=2
echo "First $n prime numbers are:"
while [ $count -lt $n ]
do
flag=0
for (( i=2; i<=num/2; i++ ))
do
if [ $((num % i)) -eq 0 ]; then
flag=1
break
fi
done
if [ $flag -eq 0 ]; then
echo -n "$num "
count=$((count + 1))
fi
num=$((num + 1))
done
Enter the value of n
1,2,3,5,13
First 1,2,3,5,13 prime numbers are:
bash: [: 1,2,3,5,13: integer expression expected

ASUS@LAPTOP-I4P5FEVA MINGW64 ~/OneDrive/Desktop/cd24021--os (main)
$
```

5. Write menu driven program for file handling activity

- **Creation of file.**
- **Write content in the file.**
- **Upend file content.**
- **Delete file content.** **Output 5 :**

```
ASUS@LAPTOP-I4P5FEVA MINGW64 ~/OneDrive/Desktop/cd24021--os (main)
$ #!/bin/bash
echo "1) Create File"
echo "2) Write Content"
echo "3) Append Content"
echo "4) Delete File Content"
echo "Enter choice:"
read ch
echo "Enter file name:"
read fname
case $ch in
1)
touch $fname
echo "File created"
;;
2)
echo "Enter content (Ctrl+D to save):"
cat > $fname
;;
3)
echo "Enter content to append (Ctrl+D to save):"
cat >> $fname
;;
4)
> $fname
echo "File content deleted"
;;
*)
echo "Invalid choice"
;;
esac
1) Create File
2) Write Content
3) Append Content
4) Delete File Content
Enter choice:
)
Enter file name:
khushabu_CD24021
Invalid choice

ASUS@LAPTOP-I4P5FEVA MINGW64 ~/OneDrive/Desktop/cd24021--os (main)
$
```